

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and maintained **Java Spring Boot** microservices, implementing REST APIs for inter-service communication and data exchange, handling approximately **500 requests per second** during peak loads.
- Designed and implemented efficient data persistence layers using **Hibernate** and **JPA**, optimizing database interactions for a core enterprise application that processed over **10,000 transactions daily**.
- Authored comprehensive unit and integration tests using **JUnit** and **Mockito** for critical Java modules, increasing code coverage by **18%** and reducing post-deployment defects.
- Contributed to the design and implementation of a new **cybersecurity** module within the Computer Science curriculum, focusing on secure coding practices and threat modelling.
- Mentored junior engineers on best practices for **software engineering**, including code review standards, version control workflows, and Agile methodologies, improving team velocity by **15%**.
- Collaborated with academic staff to define technical specifications and learning objectives for new undergraduate courses in **distributed systems**, ensuring alignment with industry demands.
- Containerised existing Java applications using **Docker** and deployed them to a managed **Kubernetes** cluster on AWS, streamlining the development-to-production workflow and enabling faster release cycles.

SKILLS

Programming Languages – Java, Python, SQL, R, TypeScript

Frameworks & Libraries – Spring Boot, Hibernate, JPA, JUnit, Mockito, pandas, NumPy, scikit-learn, Flask, FastAPI, NestJS

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes

Databases – PostgreSQL, MySQL, Oracle

Tools & Methodologies – Jenkins, Git, Jira, Confluence, Splunk, Dynatrace, Linux, Agile, Scrum

Data Analytics & BI – Power BI, Tableau, DAX, Power Query, Data Modelling, Data Warehousing, Statistical Analysis

Domain Expertise – Cybersecurity, Software Engineering, Distributed Systems, Financial Services Operations

PROJECTS

Secure Software Engineering Curriculum Module

- Developed a comprehensive curriculum module on **secure software engineering** practices, integrating concepts of threat modelling, secure coding standards, and vulnerability assessment for undergraduate students.
- Designed practical lab exercises and assignments focused on identifying and mitigating common security flaws in **Java** and Python applications, resulting in a **20% improvement** in student-identified vulnerabilities during testing.
- Created detailed documentation and teaching materials, including lecture notes and code examples, to support the delivery of the module within the Computer Science MSc program.

Distributed Systems Simulation Framework

- Engineered a simulation framework in **Java** to model and analyze various **distributed systems** architectures, including consensus algorithms and fault tolerance mechanisms.
- Implemented core components for node communication, message passing, and failure injection, allowing for the testing of system resilience under adverse conditions.
- Configured and ran simulations on a local **Docker**-based cluster, collecting performance metrics such as message latency and system throughput to evaluate different design choices.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)